Circular Motion

Motion of a body in a circular path (or circle) is called circular motion. If the speed of the body in the circular motion is constant, then the motion is called uniform circular motion. E.g. motion of the moon around the earth, stone revolved in a circle with constant speed with the help of string etc. The tangent drawn at any point on the circle gives direction of velocity at that point. Although the speed is constant in the uniform circular motion, the direction of velocity is different at different points, which produces an acceleration. Therefore, the uniform circular motion is an accelerated motion. The circular motion is a two dimensional motion.

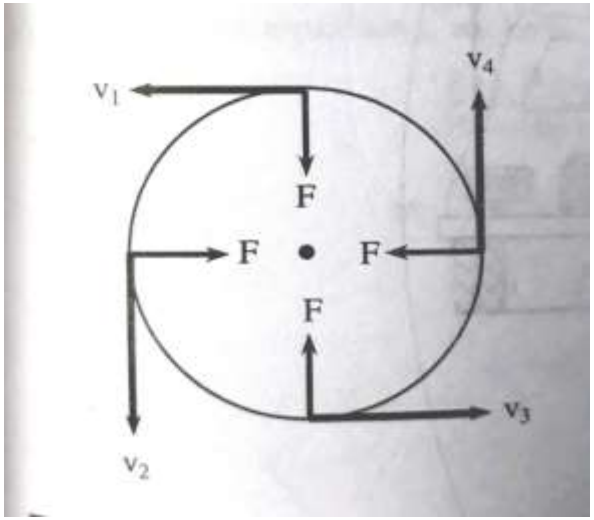


Fig: Uniform Circular Motion

Terms Related to Circular Motion

❖ *Angular displacement* (θ):

It is defined as the angle traced out by the radius vector at the center of the circular path in a given interval of time.

In figure, $< QOP = \theta$ is angular displacement in small time 't'. Its SI-unit is radian.

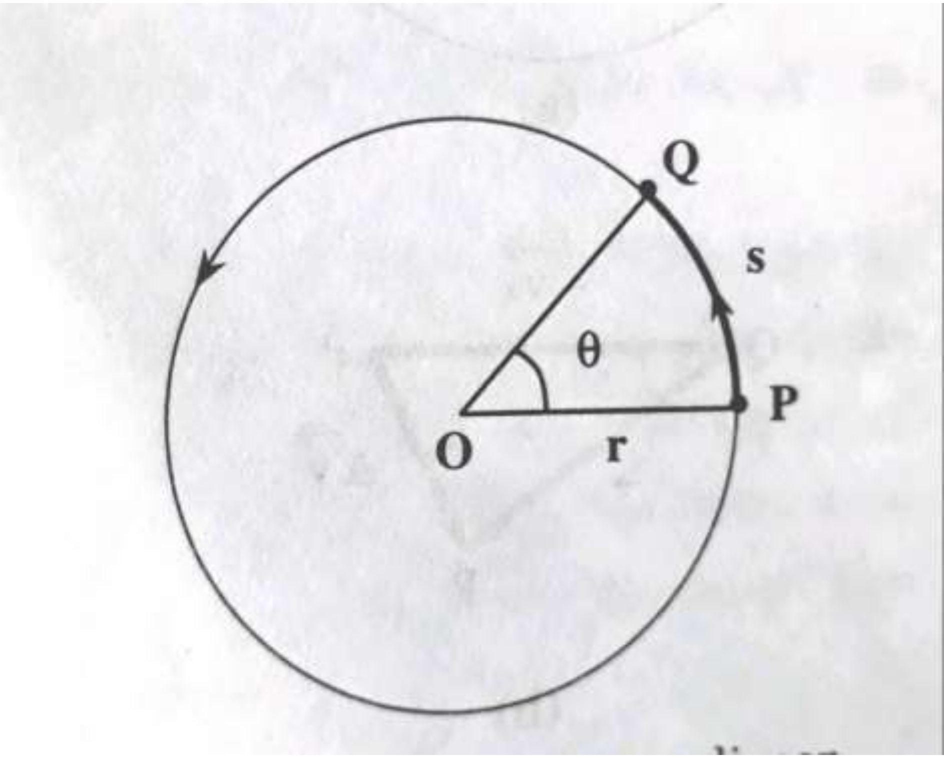


Figure: Angular Displacement

❖ Angular velocity (ω):

It is defined as the rate of change of angular displacement with respect to time. (Unit: rad/Sec)

Consider a particle moving in a circular path of radius 'r' with center 'O'. Suppose, at time t=0, the object is at point 'P'. Let, it reaches to the point 'Q_1' in time 't_1' and to the point 'Q_2' in time 't_2' as in figure. In figure, $< POQ_1 = \theta_1$ *and* $< POQ_2 = \theta_2$ are angular displacements in time 't_1' and 't_2' respectively.

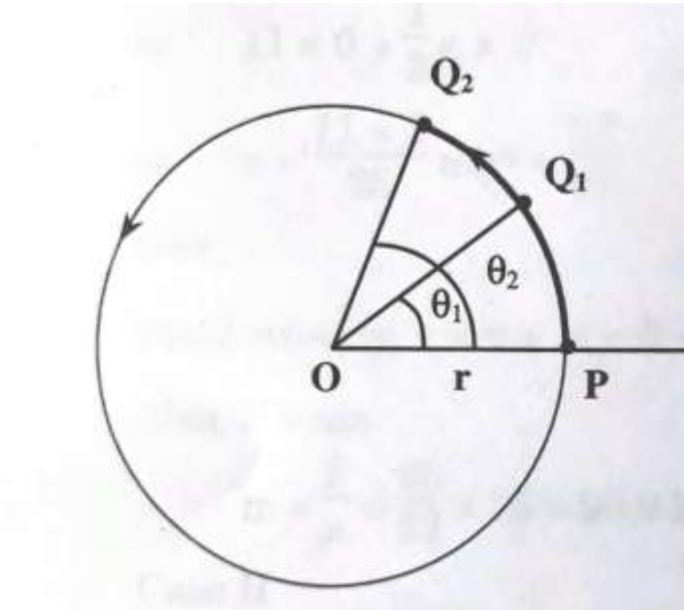


Fig: Average angular velocity between Q_1 and Q_2

Then the average angular velocity of the particle from Q_1 *to* Q_2 is given by;

$$\omega_{avg} = \frac{Angular\ Displacement}{Time\ interval} = \frac{\theta_2 - \theta_1}{t_2 - t_1}$$

$$\therefore \omega_{avg} = \frac{\Delta\theta}{\Delta t}$$

Instantaneous angular velocity (ω_{inst}):

The velocity of the particle at any instant (or at any point on the circle) is called instantaneous velocity.

$$\omega_{inst} = \lim_{\Delta t \to 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}$$

❖ Angular acceleration (α):

It is defined as the rate of change of angular velocity of a particle with respect to time. (it arises when the motion is non-uniform). Unit: rad/S^2.

$$i.e. \alpha = \frac{Change\ in\ angualr\ velocity}{Time\ interval}$$

❖ Average angular acceleration (α_{avg}):

If $\omega_1\ and\ \omega_2$ are the angular velocities of a particle in time $t_1\ and\ t_2$ respectively, then, we have;

$$\alpha_{avg} = \frac{\omega_2 - \omega_1}{t_2 - t_1} = \frac{\Delta\omega}{\Delta t}$$

❖ Instantaneous angular acceleration (α_{inst}):

It is defined as the angular acceleration of the particle at a particular instant of time during the circular motion.

$$\alpha_{inst} = \lim_{\Delta t \to 0} \frac{\Delta\omega}{\Delta t} = \frac{d\omega}{dt}$$

$$As\ \omega = \frac{d\theta}{dt}, \alpha_{inst} = \frac{d}{dt}\left(\frac{d\theta}{dt}\right)$$

$$\therefore \alpha_{inst} = \frac{d^2\theta}{dt^2}$$

❖ Time Period (T):

It is defined as the time taken by a particle to complete one revolution in a circular path. As the particle completes one revolution (i.e. angular displacement of 2π radian) in time ‘T’, we have;

$$\omega = \frac{2\pi}{T}$$

❖ Frequency (f):

It is defined as the number of revolutions completed by the particle in one second.

Unit: Hertz (Hz).

We have; $f = \frac{1}{T}$

And $\omega = \frac{2\pi}{T}$ = 2πf

❖ Relation between linear and angular velocity:

Consider a particle moving with uniform speed ‘V’ in a circular path of radius ‘r’ with center ‘O’. Let, the particle moves from initial position ‘P’ to the final position ‘Q’ in time ‘t’ such that the linear displacement along the circular path is PQ≈S and the angular displacement at the center is ‘θ’.

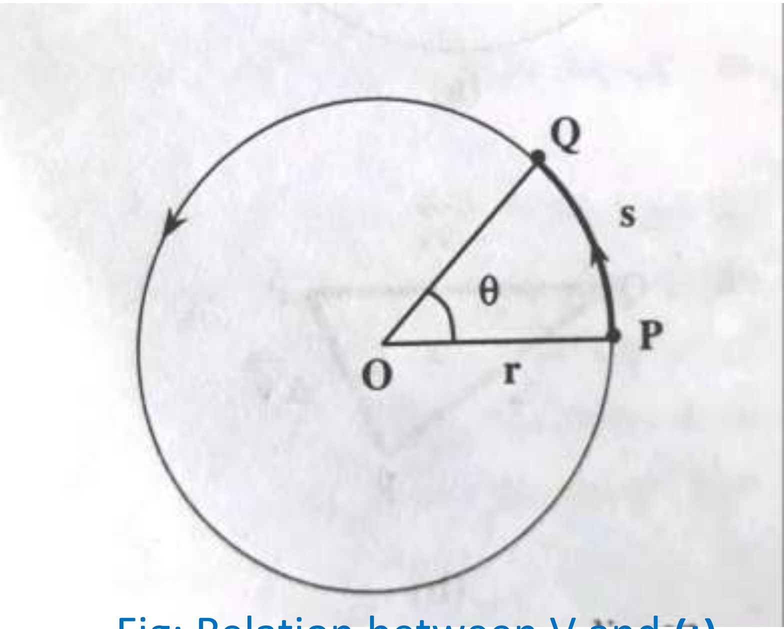


Fig: Relation between V and ω

From figure, for small angle ‘θ', *we have*;

$$\theta = \frac{S}{r}$$

$$or, S = r\theta$$

Differentiating with respect to time ‘t’, we get;

$$\frac{dS}{dt} = r\frac{d\theta}{dt}$$

$$\therefore \mathrm{v} = \mathrm{r}\omega \dots \dots \dots .. (1)$$

$where, v = \frac{dS}{dt}$ *is linear velocity and* $\omega = \frac{d\theta}{dt}$ *is angular velocity.*

Centripetal Force and Centripetal acceleration:

The force which causes a body to move in a circular path with uniform speed is called centripetal force. This force acts at every point on the circular path in such a way that it would change the direction of velocity (i.e. deflects the particle from its tangential path to the circular path) without changing the magnitude of velocity. Therefore, the force must always act at right angle to the circular path at every point.i.e.it acts along the radius vector and towards the center. According to Newton's second law, net force acting on a particle produces an acceleration and the acceleration produced by the centripetal force is called centripetal acceleration. Its direction is same as that of the centripetal force. It arises due to the change in direction of velocity of the particle in the uniform circular motion.

Consider a particle of mass ‘m’ moving with a constant speed ‘v’ in a circular path of radius ‘r’ with center at ‘O’. Let, the particle moves from the point ‘A’ to the point ‘B’ in small interval of time ‘$\Delta t'$ such that the linear displacement along the circle is ‘$\Delta l'$ and the angular displacement at the center is ‘$\Delta\theta'$ as in figure (a). Let, $\overrightarrow{V_A}$ and $\overrightarrow{V_B}$ be the velocities of the particle at the points ‘A’ and ‘B’ respectively.

In figure (b), $-\overrightarrow{V_A}$ and $\overrightarrow{V_B}$ are redrawn as the two sides of the ΔPQR, where; $\overrightarrow{\Delta V} = \overrightarrow{V_B} - \overrightarrow{V_A}$ represents the change in velocity.

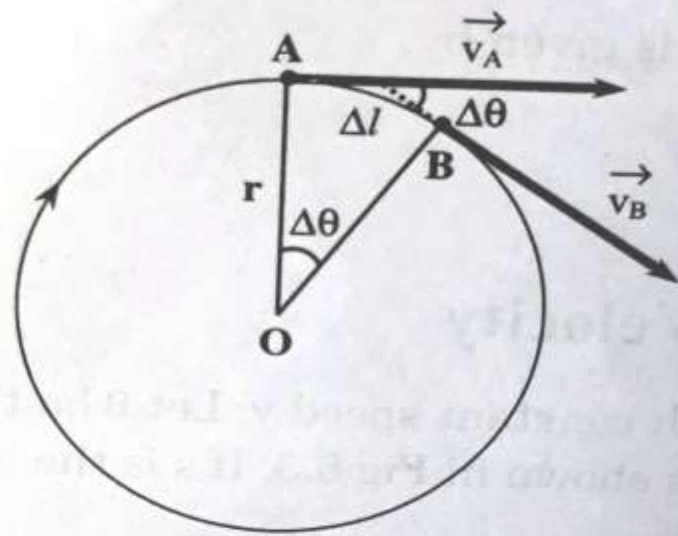
From geometry, we see that the triangles; ΔOAB and ΔPQR are similar. Therefore, from figure, we have;

$$\frac{\Delta V}{V} = \frac{\Delta l}{r}$$

$$or, \Delta V = \frac{V}{r}\,\Delta l$$

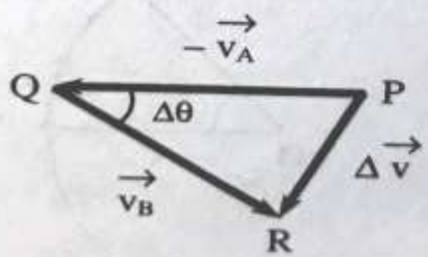
Diff. with respect to time, we get;

$$\frac{\Delta V}{\Delta t} = \frac{V}{r}\frac{\Delta l}{\Delta t} \ldots\ldots\ldots\ldots (1)$$



(a)

Fig: Uniform Circular Motion



(b) Fig:Trianle law to find $\overrightarrow{\Delta V}$

For time $\Delta t \to 0$,

$$\lim_{\Delta t \to 0} \frac{\Delta V}{\Delta t} = \mathrm{a}$$

$$and \ \lim_{\Delta t \to 0} \frac{\Delta l}{\Delta t} = V$$

Then, the equation (1) becomes;

$$a = \frac{V}{r} V$$

$$\therefore a = \frac{V^2}{r} \dots \dots \dots . (2)$$

When $\Delta t \to 0$, the point 'B' lies very close to 'A' and hence $'\Delta\theta'$ becomes very small. Then the change in velocity $\overrightarrow{\Delta V}$ becomes perpendicular to $\overrightarrow{V_A}$ and $\overrightarrow{V_B}$. Therefore, the acceleration is perpendicular to velocity and acts along the radius vector and towards the center. This acceleration is called centripetal or radial acceleration.

Again, v= $r\omega$, then the equation (2) becomes;

$$a = r\omega^2 \dots \dots \dots . (3)$$

From Newton's second law;

Centripetal force= mass x centripetal acceleration

$$\therefore F = \frac{mv^2}{r} = mr\omega^2 \dots \dots \dots \dots (4)$$

It acts along the radius vector and toward the center.

❖ Relation between Linear and radial acceleration:

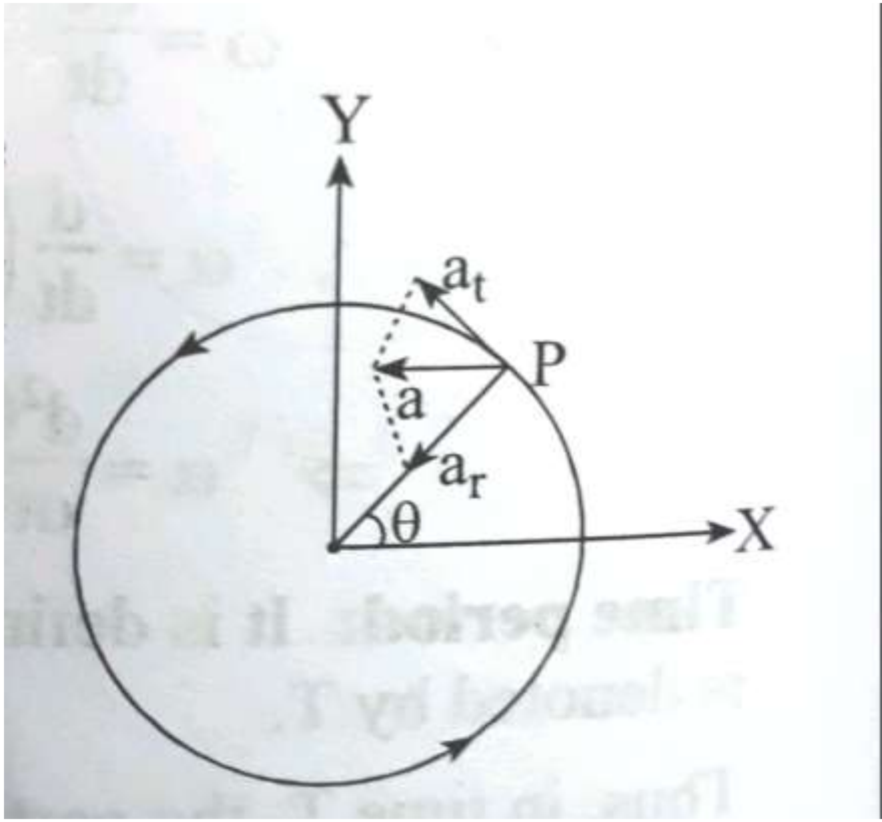
For a particle moving in a circular path with non-uniform speed, there are two types of acceleration;

(i) Linear or Tangential acceleration (a_t) along the tangential direction and

(ii) Radial or Centripetal acceleration (a_r) along the radius vector or towards the center.

Then, the resultant acceleration of the particle for the non-uniform motion is given by;

$a = \sqrt{a_t^2 + a_r^2}$

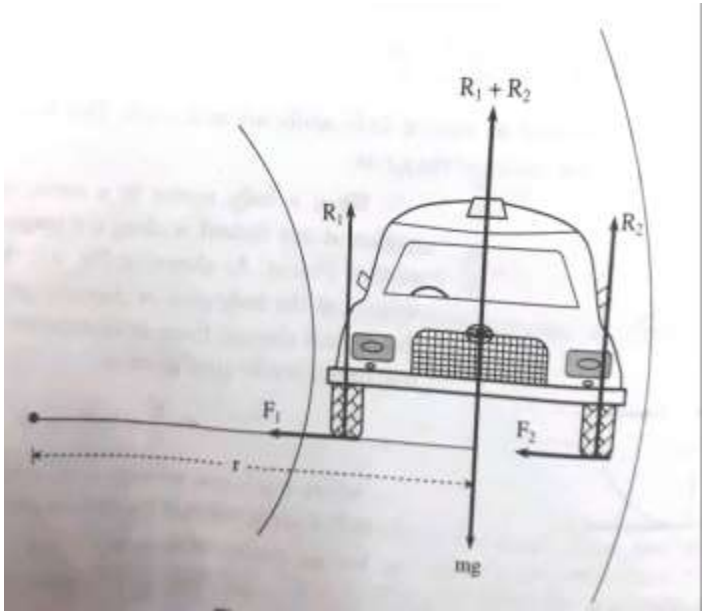


Motion of a Vehicle on a level curved road:

- When a vehicle moves around a level (or horizontal) curved path, the necessary centripetal force is provided by the frictional force between the wheels and the road, acting towards the center of the circular path.
- Suppose a vehicle of mass 'm' is moving around a curved path of radius 'r' as in figure. The weight of the vehicle '*mg*' acts vertically downwards. Let, 'R_1'*and* 'R_2' are the normal reactions of the road on the wheels which act vertically upward as in figure. Obviously, $R_1 + R_2 = mg$
- Let 'μ' be the coefficient of friction between the wheels and the road, 'F_1' and 'F_2' be the forces of friction between the wheels and the road, directed towards the center of the curved path.

$$\therefore F_1 = \mu R_1 \text{ and } F_2 = \mu R_2$$

- If 'v' be the velocity of the vehicle along the curved path, then; centripetal force = frictional force



$$\text{i.e. } \frac{mv^2}{r} = F_1 + F_2 = \mu R_1 + \mu R_2$$

$$\text{or, } \frac{mv^2}{r} = \mu(R_1 + R_2)$$

$$\text{or, } \frac{mv^2}{r} = \mu mg$$

$$\therefore v = \sqrt{\mu r g} \ldots\ldots(1)$$

This gives maximum possible velocity of the vehicle around the level curved path. If 'v' exceeds this limit, the vehicle skids out of the road.

Banking of road:

- The phenomenon of raising of outer edge of the curved road above the inner edge in order to provide necessary centripetal force for the vehicles is called Banking of Roads.
- *When a vehicle moves along horizontal curved road, necessary centripetal force is provided by the force of friction between the wheels of vehicle and the road. Such frictional force may not enough and reliable every time as it changes when road becomes icy or wet in rainy season. Further, it may not be sufficient to provide the desired speed of the vehicles and it produces wear and tear of the tires of the wheels.*
- *When the roads are banked on the curved paths, the horizontal component of the normal reaction provides the necessary centripetal force for the vehicles. Therefore, in order to provide the necessary centripetal force for the vehicles and make their motion independent(or less dependent) of the force of friction, the roads are banked at the curved paths.*

- Consider a vehicle of mass 'm' moving with constant speed 'v' around a banked curved path of radius 'r' as in figure. Let, $'\theta'$ be the angle made by the banked road with the horizontal, called angle of banking.
- The weight 'mg' of the vehicle act vertically downwards while the normal reaction 'R' acts upward by making an angle 'θ' with the vertical direction.
- The normal reaction 'R' has two components; $'RCos\theta'$ and $'RSin\theta'$.

The component $'RCos\theta'$ acts vertically upwards and balances the weight of the vehicle.

$$i.e.\ RCos\theta = mg \ldots..(1)$$

And the component $'RSin\theta'$ acts towards the center of circular path and provides the necessary centripetal force.

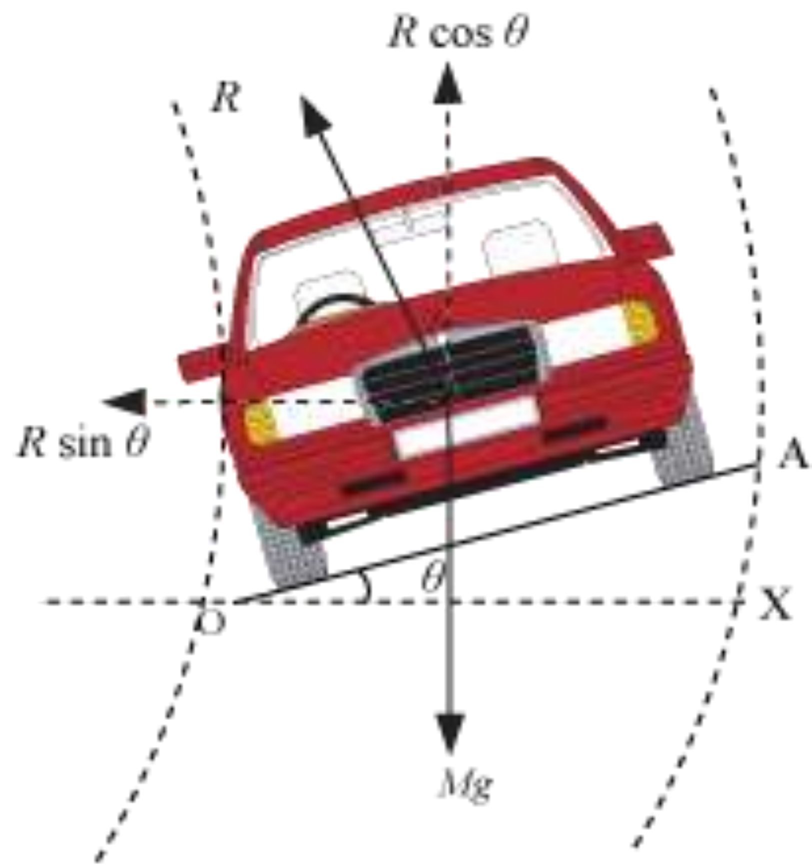
$$\text{i. e. } RSin\theta = \frac{mv^2}{r} \ldots\ldots.(2)$$

Dividing, we get;

$$Tan\theta = \frac{v^2}{rg}$$

$$\therefore \theta = \tan^{-1}\left(\frac{v^2}{rg}\right) \ldots\ldots(3)$$

$$and\ v = \sqrt{rgTan\theta} \ldots\ldots\ldots.(4)$$



Bending of a Cyclist around a curved path:

- A cyclist has to bend himself slightly towards the center of the circular track by making certain angle with the vertical direction in order to provide the necessary centripetal force.
- Consider a cyclist (cycle +man) of mass 'm' taking a turn around a circular path of radius '*r*' with velocity '*v*' *as in figure*. Suppose, he bends inwards by making an angle $'\theta'$ *with the vertical as in figure.* The weight 'mg' of the cyclist acts vertically downward while the normal reaction 'R' acts upwards by making an angle 'θ' with the vertical direction. The normal reaction 'R' has two components; $RCos\theta$ and $RSin\theta$.

The component $'R\ Cos\theta'$ acts vertically upwards and balances the weight of the cyclist.

$$R\ Cos\theta = mg\ \ldots .(1)$$

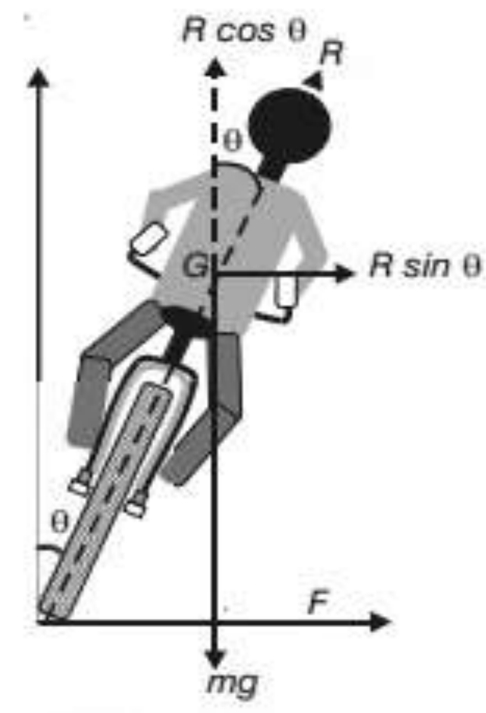
And the component $'RSin\theta'$ acts towards the center of circular path and provides the necessary centripetal force.

$$R\ Sin\theta = \frac{mv^2}{r} \ldots \ldots \ldots (2)$$

Dividing, we get;

$$Tan\theta = \frac{v^2}{rg} \ldots \ldots \ldots .(3)$$

$$\therefore\ \theta = \tan^{-1}(\frac{v^2}{rg}) \ldots \ldots .(4)$$



This gives the angle through which the cyclist has to bend while moving through a circular path of radius 'r' with speed 'v'.

Motion of Conical Pendulum

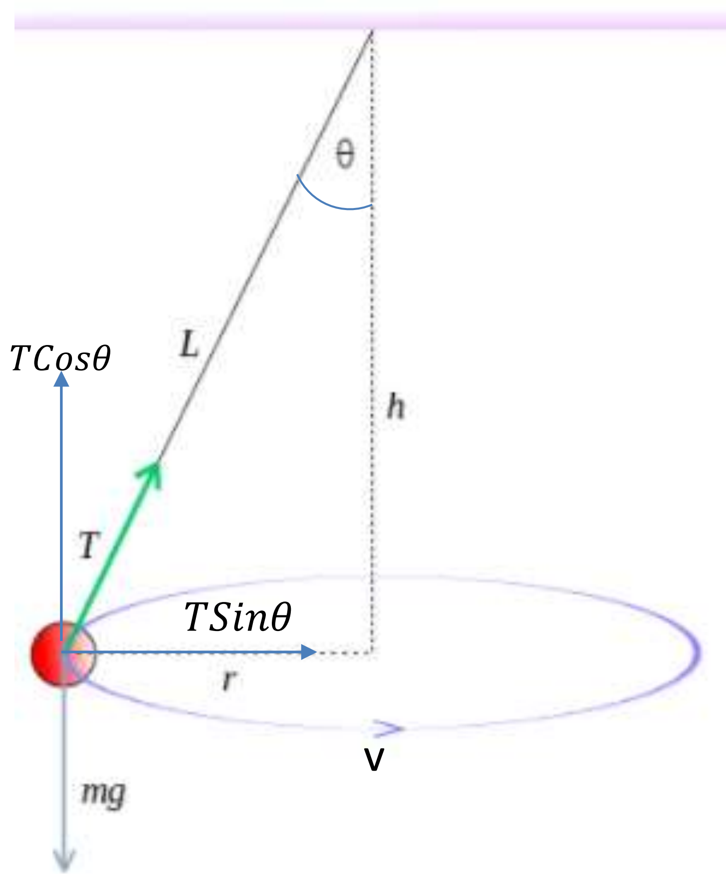
- A **conical pendulum** consists of a bob fixed at the one end of a string suspended from a rigid support where the bob moves with a constant speed in a horizontal circle such that the angle made by the string with the vertical remains constant.

- Consider a conical pendulum of length 'L' with a bob of mass 'm' suspended from a rigid support where the bob is revolving with a constant speed 'v' in a horizontal circle of radius 'r' as in figure. Let, $'\theta'$ be the angle made by the string with the vertical.

Let $'T'$ be the tension acting on the string. The tension $'T'$ has two components; $TCos\theta \ and \ TSin\theta$. The vertical component $'TCos\theta'$ balances the weight of the bob $'mg'$ and the horizontal component $'TSin\theta'$ provides the necessary centripetal force for the bob.

i.e. $TCos\theta = mg \ldots \ldots \ldots (1)$

$$and \ TSin\theta = \frac{mv^2}{r} \ldots \ldots \ldots (2)$$



Dividing, we get;

$$Tan\theta = \frac{v^2}{rg}$$

$$or, (r\omega)^2 = rgTan\theta$$

$$or, r\left(\frac{2\pi}{t}\right)^2 = gTan\theta$$

$$or, t = 2\pi\sqrt{\frac{r}{gTan\theta}} \ldots \ldots \ldots \ldots \ldots (3)$$

Again, from figure, we have;

$$Sin\theta = \frac{r}{L}$$

$$or, r = LSin\theta \ldots\ldots (4)$$

Using equation (4) in (3), we get;

$$t = 2\pi\sqrt{\left\{\frac{LSin\theta}{gTan\theta}\right\}}$$

$$\therefore t = 2\pi\sqrt{\left\{\frac{LCos\theta}{g}\right\}} \text{........(5)}$$

This is required expression for the time period for the conical pendulum.

Again, squaring and adding equations (1) and (2), we get;

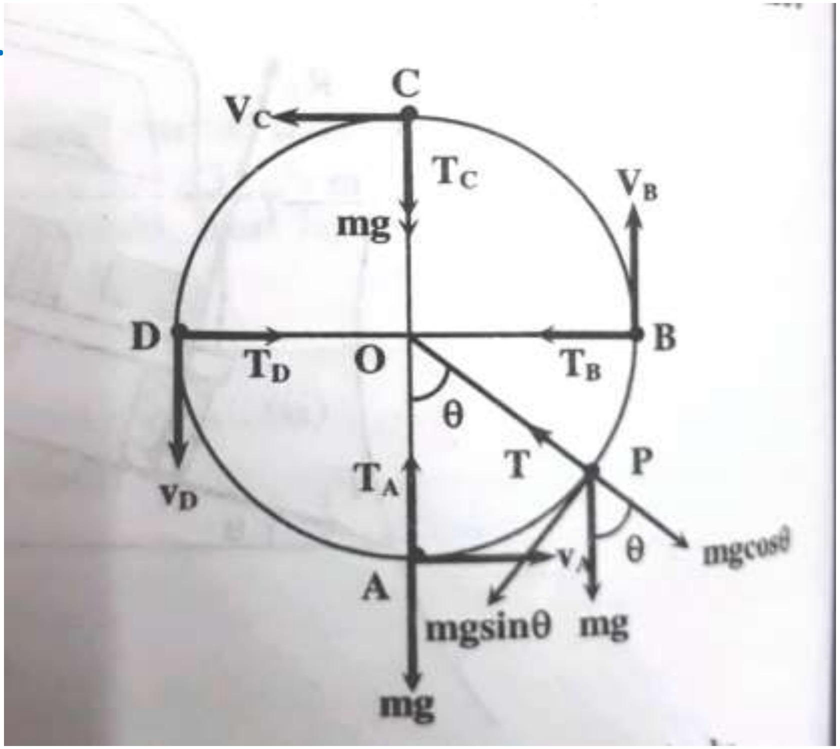
$$T^2 = (mg)^2 + \left(\frac{mv^2}{r}\right)^2$$

$$or, T^2 = (mg)^2\{1 + \left(\frac{v^2}{rg}\right)^2\}$$

$$\therefore T = mg\{1 + Tan^2\theta\}^{\frac{1}{2}} \ldots\ldots\ldots\ldots (6)$$

Motion of a particle in Vertical Circle with non-uniform velocity

- ❖ Let us consider a body of mass 'm' tied to the one end of a string which is fixed at 'O' and it is whirled in a vertical circle of radius 'r' as shown in figure.
- ❖ The motion is circular but is not uniform, since the body speeds up while coming down and slows down while going up.



- ➢ Suppose, the body is at 'P' at any instant of time t. The tension 'T' in the string always acts towards 0.
- ➢ The weight 'mg' of the body at 'P' has two components; 'mg cosθ', which acts outwards and 'mg sinθ' which acts perpendicular to the string.
- ➢ At point 'p', the force $'T - mgcos\theta'$ acts towards the center and provides the necessary centripetal force.

  Therefore;

$$T - mgcos\theta = \frac{mv^2}{r}$$

$$or, T = \frac{mv^2}{r} + mgcos\theta \ldots\ldots..(1)$$

- At the lowest point A, θ = 0º, Cos0 = 1, then;

$$T_A = \frac{mv_A^2}{r} + mg \ldots\ldots(2)(Maximum\ Tension)$$

- At the highest point C, θ = 180°, Cos180 = −1,then;

$$T_C = \frac{mv_C^2}{r} - mg\ \ \ldots\ldots..(3)(Minumum\ tension)$$

- At the intermediate points ‘B’ and ‘D’, $\theta = 90° \ or\ 270°, Cos\theta = 0$

$$T = \frac{mv^2}{r}, where, v_B = v_D = v \ldots\ldots\ldots..(4)$$

- If $T_C > 0$, then the string remains taut, while if $T_C < 0$, the string slackens and it becomes impossible to complete the motion in the vertical circle.
- If the velocity $'v_C'$ is decreased, the tension $'T_C'$ in the string also decreases, and becomes zero at a certain minimum value of the speed, called critical velocity. The critical speed is defined as the minimum speed required for the body to complete the vertical circle at the highest point.
- Therefore;

$$\frac{mv_C^2}{r} - mg = 0$$

$$or, v_c^2 = gr$$

$$\therefore v_c = \sqrt{gr} \ldots\ldots.(5)$$

- If the velocity of the body at the highest point ‘C’ is below this critical velocity, the string becomes slack and the body falls downwards instead of moving along the circular path.

- The velocity $'v_A'$ of the body at the lowest point 'A' can be obtained by using law of conservation of energy. When the body rises from A to C, i.e. through a height 2r, its potential energy increases by an amount equal to the decrease in kinetic energy.
  ∴ Potential energy at A + Kinetic energy at A = Potential energy at C + Kinetic energy at C

$$or, K.E_A = (P.E_C - P.E_A) + K.E_C$$

$$or, \frac{1}{2}mv_A^2 = mg(2r) + \frac{1}{2}mv_C^2$$

$$or, v_A^2 = 4gr + gr \qquad [\because v_C = \sqrt{gr}\,]$$

$$\therefore v_A = \sqrt{5gr} \ldots\ldots\ldots (6)$$

This is the minimum velocity required by the particle to complete the vertical circle at the lowest point.

Centrifugal Force

It is an outward force experienced by a body moving in a circular path. Its magnitude is same as that of centripetal force (i.e.$\frac{mv^2}{r}$) but its direction is opposite to that of the centripetal force. i.e. radially outward from the center.

The centrifugal force is a fictitious force (not real) as it comes into play when there is centripetal force. It arises due to the inertial property of the body moving in the circular path. Due to the inertial property, a body moves in the direction opposite to the change in velocity of the body. For example, when a moving bus suddenly stops, due to inertia of motion; the passengers inside the bus fall forward while the direction of change in velocity of the bus is in backward direction. Similarly, in the circular motion, the change in velocity takes place towards the center of the circular path and hence due to the inertial property of the body, it tends to move away from the center of the circular path.

Principle of Centrifuge

Let, a light particle of mass ‘m’ and a heavy particle of mass ‘M’ are rotated in a circle of radius ‘R’ with uniform angular velocity ‘ω’. Then, the centrifugal forces experienced by the particles are given by;

$$F_m = mR\omega^2 \dots \dots \dots (1)$$

$$and\ F_M = MR\omega^2 \dots \dots \quad (2)$$

Dividing, we get; $\frac{F_M}{F_m} = \frac{M}{m} \dots \dots \dots \dots \dots \; (3)$

As M $> m, F_M > F_m$. i.e. the centrifugal force experienced by the heavy particle is greater than the centrifugal force experienced by the light particle. Therefore, the heavier particles of the mixture move away from the axis of rotation where as the lighter particles move closer towards the axis of rotation and hence they can be separated from their mixture. This is called principle of centrifuge.

Examples:

- ❖ Cream is separated from milk in a cream separator.
- ❖ Wet clothes are dried in drying machines.
- ❖ Sugar crystals are separated from their molasses.
- ❖ Honey can be separated from the mixture honey and wax.